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**CAPSTONE PROJECT PROPOSAL**  
**ONLINE INFORMATION AND TOURIST NAVIGATION FOR KML**  
**MOUNTAIN RESORT**

In partial fulfillment for the requirements in Capstone 1 and Research Subject

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**CHAPTER I**

**INTRODUCTION**

**Background of the Study**

The rapid growth of web-based technology has transformed how people discover travel destinations and navigate unfamiliar locations. Online platforms allow resorts to reach guests far beyond their local area, providing essential information and precise directions before a traveler even leaves home.

However, many smaller leisure destinations, such as KML Mountain Resort, face significant challenges in this digital landscape. Despite offering premium amenities like panoramic mountain views and private accommodations, the resort suffers from low online visibility and difficult physical navigation. Currently, the resort's location is hard for search engines to find, and its remote nature often confuses guests. Furthermore, the lack of a centralized digital hub means the owners must manually answer every inquiry, which is inefficient for a small-scale operation.

To address these issues, modern web development offers a targeted solution. By integrating Interactive Map APIs and Search Engine Optimization (SEO), a resort can transform from a hidden location into a highly visible and easy-to-reach destination. An integrated web portal can provide real-time navigation using GPS coordinates and visual landmarks, ensuring that guests are guided directly to the resort gates without confusion.

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This capstone project proposes the development of an Online Information and Tourist Navigation System for KML Mountain Resort. The system is designed to provide precise navigation tools, a digital information hub, and increased search engine discoverability. By applying principles of web development, database management, and digital mapping, this project serves as a practical application of the knowledge gained by BSIT students to solve real-world business challenges.

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**Statement of the Problem**

KML Mountain Resort suffers from low online visibility and difficult physical navigation due to its remote location, which hinders its ability to reach its target market and results in lost potential revenue and operational inefficiencies.

1.Low Search Engine Discoverability: The resort lacks an official SEO-optimized digital platform that makes it difficult for city-based and foreign tourists to find the business through organic online searches.

2.Confusing Navigations: Because the resort is "secluded" and remote the standard GPS tools often fail to provide the precise route leading to guest frustration and difficulty in locating the entrance.

3.Manual Information Dissemination: There is no centralized digital gallery or information hub, forcing the owners to manually handle repetitive inquiries regarding room arrangements, food services, and resort facilities.

4.Limited Brand Awareness: Without a professional web presence, the resort struggles to compete with more visible destinations, limiting its ability to attract event organizers and families looking for a secure getaway.

This study proposes the development of an integrated web-based solution that combines an official information hub, interactive landmark-based navigation, and an administrative content management system. The goal is to enable small-scale leisure destinations like **KML Mountain Resort** to overcome challenges related to low online visibility and difficult physical navigation, ultimately positioning the business for long-term growth and digital competitiveness.

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This capstone project seeks to answer the following questions:

- How can the system improve discoverability to ensure that city-based and foreign tourists can easily find the resort through organic online searches?
- How can Interactive Map Integration and visual landmarks be utilized to provide precise, turn-by-turn navigation that guides guests directly to the resort gates?
- How can the system automate the dissemination of resort information to reduce the manual administrative burden of answering repetitive inquiries regarding facilities and services?
- How can a centralized digital gallery and information hub be designed to showcase the resort's premium amenities and set accurate guest expectations?
- How can the system provide a seamless user flow that transitions a potential guest from discovering the resort online to initiating live mobile navigation?
- How can the administrative dashboard be structured to allow the owners to efficiently manage content, update landmark data, and monitor user analytics?
- How can the system's user interface be designed using responsive web principles to enhance usability and ensure a consistent experience across both mobile and desktop browsers?
- How can the system be designed to handle future scalability, ensuring it can support additional features or the digital transformation of other remote small businesses?

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**Objectives of the Study**

The primary goal of this study is to design and develop an Online Information and Tourist Navigation System for KML Mountain Resort that enhances the resort's digital visibility and provides precise, landmark-based navigation for guests.

Specifically, the study aims to:

- To design a web-based portal that serves as the official digital information hub, showcasing the resort's facilities, services, and digital gallery to set guest expectations.
- To implement an Interactive Map Integration using GPS coordinates and visual landmarks to guide tourists directly to the resort gates through their mobile GPS applications.
- To streamline the user flow for potential guests by providing a seamless transition from discovering the resort online to initiating turn-by-turn navigation.
- To reduce manual administrative tasks for the owners by automating the dissemination of resort information and frequently asked questions.



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- To integrate Search Engine Optimization (SEO) strategies into the web portal's architecture to improve organic search rankings and increase the resort's online discoverability for local and foreign tourists.
- To develop a secure administrative dashboard that allows the resort owner to update room rates, manage the photo gallery, and modify landmark information in real-time.
- To utilize landmark-based guidance within the mapping module to provide visual cues (such as specific bridges or trees) that compensate for the inaccuracies of standard GPS tools in remote areas.

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### **Significance of the Study**

The proposed **Online Information and Tourist Navigation System** is designed to address the operational and visibility challenges commonly faced by small-scale leisure destinations in remote areas. As the demand for secluded and nature-based travel continues to grow, enterprises like **KML Mountain Resort** are under increasing pressure to modernize their digital presence and provide precise navigation to maintain competitiveness. This study presents a comprehensive digital solution that enhances business discoverability, minimizes traveler frustration, and supports sustainable growth for hidden local destinations.

At its core, the system centralizes the resort's information—from high-quality visual galleries of amenities to automated answers for frequently asked questions. This integration reduces the manual administrative workload for owners, eliminates the need for repetitive individual inquiry handling, and ensures that potential guests receive consistent information. By improving the reliability of the resort's digital "front door," the system helps lower operational inefficiencies and increases the likelihood of bookings.

A vital component of this project is the **Interactive Map Integration**, which utilizes GPS coordinates and visual landmarks to provide real-time, turn-by-turn guidance. This feature addresses the common failure of standard GPS tools in remote mountain terrains, ensuring guests are guided directly to the resort gates. This not only improves guest satisfaction but also reduces travel anxiety associated with navigating "secluded" locations.

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The inclusion of an **Administrative Dashboard** enables the owners to manage content dynamically, updating room details or landmark descriptions as the resort evolves. This ensures that the resort's digital presence remains accurate and professional, fostering trust with families and event organizers. Furthermore, the system's SEO-optimized architecture ensures that the resort becomes visible to city-based and foreign tourists searching for unique mountain getaways.

Ultimately, this project represents a strategic and future-ready solution for small-scale resorts seeking to embrace digital transformation. By integrating automation, precise mapping, and customer-focused information tools into one platform, the system enhances operational reach and promotes local tourism. This project serves as a reference model for other remote businesses in the Philippines aiming to modernize their processes and strengthen their brand through technology-driven innovation.

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### **Scope and Limitations of the Study**

This study focuses on developing and implementing the Online Information and Tourist Navigation System—a centralized platform designed to improve the digital visibility and physical reach of KML Mountain Resort. The system integrates an official information hub, high-quality digital galleries, and interactive landmark-based navigation to minimize traveler confusion, streamline administrative workflows, and enhance the overall guest experience.

The system is intended for both end-users (tourists) and administrators (resort owners), enabling guests to browse facilities and initiate live navigation while allowing management to update resort data and monitor analytics. The scope is specifically limited to the operations of KML Mountain Resort and serves as a prototype for similar "hidden" leisure destinations transitioning from manual inquiries to automated digital platforms. The project prioritizes navigation accuracy, information accessibility, and responsive web usability over complex financial integrations or third-party booking systems.

However, the study recognizes the following limitations:

- **Financial and Resource Constraints** – As the target is a small-scale resort, immediate adoption of advanced features such as real-time room inventory synchronization or integrated online payment processing is restricted. The project focuses on essential informational and navigational functions while leaving a modular framework for future upgrades.

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- **Location-Specific Configuration** – While the system is optimized for mountain terrain and landmark-based navigation, adapting it to urban environments or different business models would require significant geographic and structural adjustments.
- **User Adoption and Digital Literacy** – Since the system replaces manual information dissemination, resort staff and some guests may initially struggle with the digital workflow. To ensure smooth adoption, the user interface is designed for simplicity to minimize training time and promote efficient interaction.
- **Connectivity and GPS Dependence** – Since the system relies on real-time map API integration and coordinate-based tracking, stable internet connectivity and GPS signal strength are essential for optimal performance. Inconsistent mobile signals in remote mountain areas may experience delays in loading interactive map data.
- **External Mapping Limitations** – The accuracy of the navigation component is subject to the limitations of third-party mapping services and local network availability. While landmark-based cues are provided to assist guests, the system cannot control for sudden physical changes in the environment or telecom coverage gaps during peak travel periods.

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## **Definition of Terms**

For a clearer understanding of this study, the following terms are defined:

- **Administrative Dashboard** – Operationally, this refers to the secure back-end web interface where the resort owner, Michelle Tingson Miranda, can manage the photo gallery, update landmark descriptions, and view user analytics.
- **Digital Information Hub** – Operationally, this refers to the centralized section of the web portal that contains the resort's history, list of amenities, room rates, and food service details, aimed at reducing manual inquiries.
- **GPS (Global Positioning System)** – Lexically, a satellite-based radionavigation system. Operationally, it is the technology used by the system to acquire the real-time geographic coordinates of the guest to provide accurate navigation.
- **Interactive Map Integration** – Operationally, this refers to the embedding of digital maps (such as OpenStreetMap or Google Maps API) into the resort's portal to provide live, visual route guidance.
- **Landmark-Based Guidance** – Operationally, this refers to the use of specific physical markers (e.g., bridges, specialized trees, or unique signage) within the navigation system to help guests identify the correct path in areas where digital maps are imprecise.

- **Online Visibility** – Lexically, the presence of a brand or business in consumer search results. Operationally, it refers to the resort's ability to be found by potential tourists through search engines like Google.

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- **Responsive Web Design** – Lexically, an approach to web design that makes web pages render well on a variety of devices and window or screen sizes. Operationally, it ensures the KML portal is fully functional on both traveler's mobile phones and desktop computers.
- **SEO (Search Engine Optimization)** the process of improving the quality and quantity of website traffic from search engines. Operationally, it refers to the strategies applied to the KML portal to ensure it ranks high when tourists search for "mountain resorts in Iloilo."
- **User Flow** the path taken by a prototypical user on a website to complete a task. Operationally, it refers to the seamless journey of a guest from viewing the resort's gallery to starting the live navigation to the destination.
- **Web-Based Portal** a specially designed website that brings information from diverse sources together. Operationally, it is the primary software solution developed for this capstone project which hosts all information and navigation tools for the resort.

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**CHAPTER II**  
**REVIEW OF RELATED LITERATURE AND STUDIES**

**Foreign Studies**

**Web-Based Information Systems and Global Tourism Trends**

The study by Halkiopoulos and Giotopoulos (2022), titled *"Tourism's use of web-based information systems and the influence of tourism trends,"* examines the critical intersection between modern information technology and shifting traveler behaviors. The authors argue that the survival of tourism destinations in a post-digital era depends on their ability to utilize web-based information systems (WIS) to adapt to global trends, such as the increasing demand for "independent travel" and "nature-based tourism."

Halkiopoulos and Giotopoulos further emphasize that web-based systems are no longer just static brochures; they are dynamic tools that influence a traveler's decision-making process. For small-scale resort, this supports the development of a digital gallery and information hub that does more than list prices—it creates a digital "experience" that builds trust and excitement. By using a web-based system to present clear, high-quality information, the resort can effectively appeal to modern tourists who rely heavily on digital content to evaluate the quality and safety of a destination before visiting.

Moreover, the authors highlight the role of digital systems in managing "location-based information" to satisfy the needs of the modern traveler. They note that today's tourists expect seamless access to navigation and spatial data. By



providing precise navigation tools, the resort addresses the modern trend of "self-navigating" tourists who prefer using their mobile devices to find destinations without needing a physical guide or local assistance.

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Additionally, the study discusses how web-based information systems provide a competitive edge to small tourism enterprises by bridging the "visibility gap." Implementing an integrated system for KML Mountain Resort aligns with this concept by leveraging SEO and digital mapping to ensure the resort is visible to international and city-based travelers. The system allows a small, provincial business to appear as professional and accessible as larger, more established destinations.

The findings by Halkiopoulos and Giotopoulos (2022) reinforce the necessity of the KML Mountain Resort system. Their research confirms that a well-designed web-based information system is an essential strategic tool for small resorts, as it enhances operational efficiency, improves digital discoverability, and allows the business to capitalize on modern tourism trends through technology-driven solutions.

**Reference:** Halkiopoulos, C., & Giotopoulos, K. (2022, April). Tourism's use of web-based information systems and the influence of tourism trends. In *Transcending Borders in Tourism Through Innovation and Cultural Heritage: 8th International Conference, IACuDiT, Hydra, Greece, 2021* (pp. 407-426). Cham: Springer International Publishing.

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**Mapping Tourist Information Flow and Cultural Influence**

The study by **Park and Stepchenkova (2023)**, titled *"Invisible power of culture: Mapping tourist information flow of national DMO websites,"* investigates the underlying structures of how information is organized and disseminated through destination management organization (DMO) websites. Published in the *Journal of Travel Research*, the study highlights that the "flow" of information—how easily a user can move from initial discovery to specific logistical details—is a critical determinant of a destination's digital success.

Furthermore, the authors discuss the concept of "mapping" information to align with the mental models of tourists. Their research suggests that travelers seek a seamless transition from "inspiration" (what a place looks like) to "navigation" (how to get there). This finding directly validates the integration of specialized navigation modules and landmark-based guidance within a tourism system. According to the study, when a digital platform provides a clear, map-like progression of information, it significantly reduces the perceived complexity of visiting remote or unfamiliar locations.

Overall, the findings by Park and Stepchenkova (2023) reinforce the necessity of an integrated navigation and information portal. Their work confirms that by carefully mapping the flow of data—from search engine visibility to precise, turn-by-turn guidance—tourism operators can improve their operational reach, enhance traveler confidence, and create a more efficient digital gateway for their business.

**Reference:** Park, H., & Stepchenkova, S. (2023). Invisible power of culture: Mapping tourist information flow of national DMO websites. *Journal of Travel Research*, 62(4), 753-767.

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**Online Tourist Information Search Strategies**

The study by Zarezadeh, Benckendorff, and Gretzel (2023), titled "*Online tourist information search strategies*," provides a comprehensive analysis of the evolving behaviors and methodologies tourists use to find destination data in the digital age. Published in *Tourism Management Perspectives*, the research identifies that modern travelers no longer rely on a single source of information but instead employ complex, multi-channel search strategies. This concept is highly relevant to the development of integrated tourism platforms, as it emphasizes that a destination's digital presence must be optimized across various touchpoints—including search engines, official websites, and interactive tools—to capture the attention of potential guests.

Zarezadeh et al. highlight that the effectiveness of a tourist's search is heavily influenced by the "findability" and "accessibility" of the content. In the context of remote or small-scale destinations, this research supports the implementation of Search Engine Optimization (SEO) and centralized information hubs. By providing a structured and easily discoverable official portal, a business can intervene in the user's search journey, moving them from a broad organic search to a specific, action-oriented interaction with the destination's own platform.

Overall, the research by Zarezadeh, Benckendorff, and Gretzel (2023) underscores the strategic importance of an integrated digital solution. Their findings confirm that by understanding and catering to the online search strategies of modern tourists, destinations can improve their visibility, streamline the guest's transition from searcher to visitor, and foster a more seamless and satisfying travel experience.

**Reference:** Zarezadeh, Z. Z., Benckendorff, P., & Gretzel, U. (2023). Online tourist information search strategies. *Tourism Management Perspectives*, 48, 101140.

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**Artificial Intelligence in Tourism and Hospitality: The Tourists' Perspective**

The study by **Sousa, Cardoso, and Dias (2024)**, titled "*The use of artificial intelligence systems in tourism and hospitality: The tourists' perspective*," explores how modern travelers perceive and interact with intelligent digital systems within the hospitality sector. Published in *Administrative Sciences*, the research highlights that tourists increasingly value efficiency, personalization, and rapid information retrieval—attributes often provided by automated and AI-driven platforms. This study is highly relevant to the development of integrated tourism systems, as it provides a behavioral foundation for how automation and intelligent data management can enhance the "digital relationship" between a destination and its guests.

Sousa et al. emphasize that the integration of intelligent systems into tourism platforms significantly reduces the "wait time" for information. For small-scale or remote destinations, this research supports the transition from manual, owner-operated inquiry handling to an automated digital information hub. By implementing a system that can provide instant answers regarding facility availability, room rates, and services, a destination aligns itself with the expectations of modern travelers who prioritize immediate responsiveness and 24/7 accessibility.

Overall, the study by Sousa, Cardoso, and Dias (2024) confirms that the digitalization and automation of tourism services are essential for meeting the demands of the contemporary market. Their findings support the development of integrated navigation and information portals as a strategic means to improve operational efficiency, build traveler trust, and provide a high-quality,

technology-supported guest experience from the initial search phase to final arrival.

**Reference:** Sousa, A. E., Cardoso, P., & Dias, F. (2024). The use of artificial intelligence systems in tourism and hospitality: The tourists' perspective. *Administrative Sciences*, 14(8), 165.

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**Development of ICT: From e-Tourism to Smart Tourism**

The study by Leung (2022), titled *"Development of information and communication technology: from e-tourism to smart tourism,"* provides a comprehensive historical and technical overview of how Information and Communication Technology (ICT) has reshaped the tourism landscape. Published in the *Handbook of e-Tourism*, the research details the evolution from basic "e-tourism" (static websites and online brochures) to "smart tourism," which utilizes interconnected systems, real-time data, and mobile integration to create a seamless travel ecosystem. This research is highly relevant to the development of integrated tourism platforms, as it positions the transition from manual information sharing to automated, navigation-ready portals as an essential step in modernizing any tourism enterprise.

Leung emphasizes that the hallmark of smart tourism is the "interconnection" of information. For small-scale or provincial destinations, this supports the implementation of a centralized digital information hub that does more than just display text. By integrating dynamic galleries, service details, and location data into one interface, a destination can move away from fragmented, manual inquiry management and toward a "smart" operational model that provides immediate, high-quality information to potential guests.

A significant portion of the study focuses on the role of Mobile-Centric Services and Ubiquitous Connectivity. Leung argues that modern travelers expect "on-site" support, meaning they need access to information and navigation tools while they are physically in transit. This directly validates the integration of

mobile-responsive design and Interactive Map APIs within a tourism system. By providing precise, real-time guidance and landmark-based routing, a system fulfills the "smart tourism" requirement of supporting the traveler throughout their entire journey—not just during the initial booking phase.

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The research by Leung (2022) serves as a theoretical bridge for the development of integrated navigation and information systems. Their findings confirm that by embracing the principles of smart tourism—specifically through centralized data, mobile accessibility, and interactive navigation—tourism destinations can significantly enhance their operational efficiency, bridge the visibility gap, and provide a superior, technology-supported experience for modern travelers.

**Reference:** Leung, R. (2022). Development of information and communication technology: from e-tourism to smart tourism. *Handbook of e-Tourism*, 23-55.

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**Local Studies**

**Booking and Reservation Systems and Location-Based Services**

According to Gosela and Encarnacion (2024) explore the integration of booking systems and location-based services (LBS) to create unified applications that support sustainable tourism networks. In their research, *"Booking and reservation system: A unified application using location-based services for sustainable tourism networks,"* the authors argue that modern tourism platforms must go beyond simple information display by integrating real-time geographic data to enhance the traveler's experience.

Gosela and Encarnacion further emphasize that unified applications improve operational efficiency for small-scale businesses by centralizing guest interactions and location tracking into a single digital interface. In the context of a remote resort, such a system can streamline the process of handling inquiries while simultaneously providing guests with the exact coordinates and landmark-based routes needed to reach the destination. By reducing the complexity of navigation and automating information dissemination, the system minimizes traveler frustration and enhances the professional image of the resort.

Moreover, the authors highlight the importance of "Location-Based Services" (LBS) in promoting sustainable tourism. These systems enable administrators to monitor how users interact with the map and which routes are

most frequently accessed. This capability is essential for managing a resort effectively, as it allows the owner to understand guest behavior, update landmark descriptions based on real-time feedback, and support long-term strategic planning for the resort's accessibility.

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Additionally, Gosela and Encarnacion discuss how these digital networks contribute to a competitive advantage for provincial destinations. Implementing a unified system for resorts aligns with this idea by leveraging digital mapping and reservation features to meet the needs of modern tourists who require reliable, technology-supported navigation when exploring unfamiliar territories.

Overall, the findings presented by Gosela and Encarnacion (2024) support the necessity of developing an integrated navigation and information portal for KML Mountain Resort, as such a system enhances business discoverability, improves guest satisfaction through precise guidance, and enables data-driven management for the resort owners.

**Reference:** Gosela, R. R. U., & Encarnacion, R. E. (2024). Booking and reservation system: A unified application using location-based services for sustainable tourism networks. *International Journal of Advanced Research in Science Communication and Technology*, 4(1), 739–749. ISSN:2581-9429.

**Integrated Simulation and Management Systems in Tourism Education and Operations**

The study by Altar et al. (2024), titled "*An Integrated Intranet-Based Simulation System for College of Hospitality and Tourism Management of EARIST*"



*Manila,"* focuses on the development of a centralized digital environment designed to simulate and manage hospitality and tourism operations. Published by IEEE, the research emphasizes how integrating multiple functional modules into a single system—such as guest handling, resource management, and information dissemination—creates a more efficient and organized environment for both administrators and end-users.

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Altar et al. highlight that integrated systems improve operational efficiency by providing a structured workflow that reduces the reliance on manual, decentralized processes. For a small-scale enterprise, this supports the transition from manual inquiry handling to an automated "Digital Information Hub." By centralizing resort data, the system ensures that information remains consistent and accessible, mirroring the study's goal of streamlining hospitality management through technology to minimize human error and operational delays.

Furthermore, the authors discuss the importance of using simulation-style interfaces to improve user familiarity and ease of use. In the context of the KML project, this aligns with the objective of creating an intuitive Administrative Dashboard. Such a dashboard allows the resort owner to simulate or manage the "guest journey"—from updating the photo gallery to refining landmark-based navigation data—ensuring that the digital representation of the resort is always accurate and ready for potential visitors.

The study by Altar et al. (2024) validates the structural approach of the KML Mountain Resort project. It supports the necessity of developing a system that integrates administrative management with user-facing information tools, ultimately enhancing the resort's operational reach, improving guest satisfaction, and providing a sustainable digital framework for growth.

**Reference:** Altar, J. R. B., Dellova, C. A. P., Resco, A. M., Bartiquin, A. M., Dinaga, M. R. D., & Costales, J. A. (2024, November). An Integrated Intranet-Based Simulation System for College of Hospitality and Tourism Management of EARIST Manila. In *2024 3rd International Conference on Computer Applications Technology (CCAT)* (pp. 55-60). IEEE.

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**Impact and Identities in the Linguistic Landscape of Tourist Destinations**

The study by Sieras (2024), titled *"Impact and Identities as Revealed in Tourists' Perceptions of the Linguistic Landscape in Tourist Destinations,"* examines how the physical signs, languages, and visual cues (the "linguistic landscape") within a destination influence traveler perceptions and behaviors. Published in the *International Journal of Language and Literary Studies*, the research highlights that the way information is visually communicated—through signage, digital interfaces, and landmarks—plays a critical role in shaping a tourist's sense of security, orientation, and connection to a place. This study is highly relevant to the development of integrated tourism platforms, particularly those utilizing **landmark-based navigation**, as it provides a sociolinguistic foundation for how visual markers help guests "read" and navigate an environment.

Sieras emphasizes that clear and accessible linguistic cues are essential for "wayfinding" in unfamiliar territories. For remote or secluded destinations, this research supports the implementation of navigation systems that go beyond mere coordinates to include descriptive, landmark-based guidance. By integrating specific visual cues (such as descriptions of local bridges, unique trees, or specific signs) into a digital navigation portal, a system aligns with the traveler's natural tendency to look for familiar linguistic or physical "anchors" to confirm they are on

the right path. This reduces the cognitive stress of navigating "silent" or poorly marked rural routes.

The research highlights that a lack of clear informational cues can lead to a sense of exclusion or frustration among travelers. Implementing an integrated system that combines search engine discoverability with precise, landmark-oriented instructions directly addresses this challenge. The study suggests that when a destination takes control of its "informational landscape"—whether physically or digitally—it significantly improves the guest experience and encourages positive word-of-mouth.

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The findings by Sieras (2024) provide a unique perspective on the necessity of integrated navigation systems. The research confirms that by strategically using landmarks and clear digital communication, tourism destinations can improve their physical reachability, strengthen their brand identity, and ensure that travelers feel guided and secure throughout their journey.

**Reference:** Sieras, S. G. (2024). Impact and Identities as Revealed in Tourists' Perceptions of the Linguistic Landscape in Tourist Destinations. *International Journal of Language and Literary Studies*, 6(1), 375-392.

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**Related Literature**

**Design and Implementation Frameworks for Resort Management Systems**

The study by **Yazdeen and Zeebaree (2022)**, titled "*Comprehensive survey for designing and implementing web-based tourist resorts and places management systems*," provides an extensive technical overview of the architectural requirements and features necessary for modern tourism platforms. Published in the *Academic Journal of Nawroz University*, the research categorizes the essential components of a resort management system, ranging from front-end user interfaces to back-end database management. This study is highly relevant to the development of integrated tourism portals, as it establishes a standardized framework for building systems that effectively manage location-based data and guest interactions.

Yazdeen and Zeebaree emphasize that the primary goal of a web-based management system is to bridge the communication gap between the destination and the tourist. They argue that a successful system must provide high accessibility and real-time information updates. For small-scale or remote destinations, this supports the implementation of a centralized digital hub that automates the

dissemination of facility details and service rates. By adopting the structured design principles outlined in this survey, a tourism enterprise can transition from manual, error-prone communication to a streamlined digital workflow that enhances operational efficiency.

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Furthermore, the authors highlight the critical role of Geographic Information Systems (GIS) and mapping integration in modern resort management. Their survey reveals that navigation features are no longer secondary but are integral to the user's journey. This find supports the integration of Interactive Map APIs and specialized routing tools within a tourism system. According to the study, providing accurate spatial data and visual cues significantly reduces the "lost traveler" phenomenon, which is a common challenge for businesses located in secluded or complex terrains

Overall, the comprehensive survey by Yazdeen and Zeebaree (2022) serves as a technical roadmap for the development of integrated tourism systems. Their findings confirm that by combining user-centric design with advanced mapping and automated management tools, a tourism destination can significantly improve its online visibility, optimize its internal processes, and provide a superior navigational experience for its guests.

**Reference:** Yazdeen, A. A., & Zeebaree, S. R. (2022). Comprehensive survey for designing and implementing web-based tourist resorts and places management systems. *Academic Journal of Nawroz University (AJNU)*, 11(3), 113-132.

**Digitalizing Tourism and Interactive Navigation**

The study by Bashir, Asghar, Ashfaq, Raza, and Zahoor (2024), titled "*Digitalizing tourism and interactive navigation: A case study of Pakistan*," examines the transformative impact of digital tools on the accessibility and promotion of regional tourism. Published in the *Journal of Asian Development Studies*, the research highlights how interactive navigation systems and digital platforms can mitigate the "geographical isolation" of remote tourist spots. This study is highly relevant to the development of integrated tourism portals, as it provides a practical framework for using technology to overcome the physical and informational barriers associated with secluded or difficult-to-reach locations.

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Bashir et al. emphasize that the "digitalization of information" is a prerequisite for modern tourism growth. Their research demonstrates that providing a centralized platform for information and route guidance significantly increases the confidence of both domestic and international travelers. In the context of remote destinations, this supports the implementation of a digital information hub. By moving away from manual inquiry management and toward an automated digital presence, tourism enterprises can ensure that accurate data regarding facilities and services is available to potential guests 24/7, thereby enhancing operational efficiency.

A central theme of the study is the role of Interactive Navigation in enhancing the guest experience. The authors argue that standard mapping tools often lack the granular detail required for navigating complex rural or mountainous terrains. Their case study findings advocate for specialized navigation interfaces that integrate real-time tracking with localized data. This directly validates the integration of Interactive Map APIs and landmark-based guidance within a tourism system. By providing specific visual cues and precise coordinates, a system can guide tourists through "dead zones" or confusing intersections where traditional GPS might fail.

Overall, the research by Bashir et al. (2024) provides a contemporary academic justification for the development of integrated navigation and information portals. Their findings confirm that by combining interactive mapping with centralized digital management, tourism destinations can improve their discoverability, streamline traveler logistics, and foster sustainable growth through technological innovation.

**Reference:** Bashir, M., Asghar, S., Ashfaq, M., Raza, S. S., & Zahoor, S. (2024). Digitalizing tourism and interactive navigation: A case study of Pakistan. *Journal of Asian Development Studies*, 13(1), 547-566.

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**Synthesis of the Reviewed Literature**

The reviewed foreign and local studies demonstrate that integrated web systems significantly improve business visibility and guest navigation. However, limited research focuses specifically on boutique mountain resorts that require a combination of landmark-based navigation and administrative information hubs. This study addresses this gap by integrating GPS-based tourist navigation with web application development, aligning with current technological and academic trends.

The reviewed literature collectively emphasizes the critical role of Integrated Information Systems and Interactive Navigation in the modernization of the tourism industry. By analyzing the various academic perspectives provided, the synthesis focuses on four core pillars: the shift toward smart unified platforms, the necessity of location-based services (LBS), the psychology of online information search, and the operational impact of digital transformation on small-scale enterprises.

The synthesis of these sources confirms that an **Online Information and Tourist Navigation System** is not merely a marketing tool but a vital operational infrastructure. By combining unified information hubs (Laudon; Leung), interactive landmark-based navigation (Gosela; Bashir; Sieras), and user-centric mobile design (Chaffey; Sousa), small-scale destinations can overcome geographical barriers. The research collectively supports the development of such a system as a means to increase visibility, streamline guest logistics, and ensure sustainable growth in a post-digital tourism economy.

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**CHAPTER III**  
**METHODOLOGY**

**Research Design**

This study employs a developmental research design, focusing on the creation, implementation, and evaluation of the KML Mountain Resort Online Information and Tourist Navigation System. The developmental approach is appropriate for this project as it allows for the systematic design and iterative refinement of the system based on real-world feedback regarding the resort's specific navigational and promotional needs.

The research process begins with an in-depth analysis of the current challenges faced by KML Mountain Resort, particularly concerning guest



navigation and the accessibility of information. This is followed by systematic planning, design, development, and testing phases. Iterative evaluation is conducted at each stage to ensure that the digital solution effectively addresses identified operational hurdles, such as the hidden nature of the resort's location, confusing landmark-based routes, manual inquiry management, and low search engine discoverability. By utilizing this design, the proponents can ensure the final system provides a seamless and reliable experience for both the resort management and its visitors.

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**System Development Methodology**

The Agile Software Development Life Cycle (SDLC) model is adopted due to its flexibility, adaptability to changing requirements, and emphasis on stakeholder collaboration. Agile is particularly suitable for the development of the Online Information and Tourist Navigation System, where navigation accuracy, user interface preferences, and real-time data integration requirements may evolve during the development process based on field testing and user feedback.



Breakdown of Phases for the KML System:

**Requirements:** Gathering data on resort landmarks, service rates, and guest pain points regarding navigation.

**Design:** Creating wireframes for the mobile-responsive information hub and mapping out the landmark-based routing logic.

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**Development:** Coding the functional modules, including the Interactive Map API integration and the Administrative Dashboard.

**Testing:** Conducting field tests to ensure the coordinates and visual landmarks accurately guide users to the resort.

**Deployment:** Launching the web-based system for live use by potential tourists and resort management.

**Review:** Collecting feedback from actual users to refine map markers or information displays in the next iteration.

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**System Architecture**

Online Information and Tourist Navigation System for KML Mountain Resort is designed to transform the resort's daily operations into a more organized, efficient, and digital-friendly ecosystem. The system is specifically engineered to solve the persistent challenges of geographical isolation, manual inquiry handling, and the difficulty guests face when navigating via traditional landmark-based

instructions. By transitioning from a fragmented manual process to a centralized digital platform, the resort ensures that all navigational data and service information are kept in a secure, accessible, and organized environment.

The Online Information and Tourist Navigation System for KML Mountain Resort utilizes a Three-Tier Architecture to ensure a structured, scalable, and organized flow of data. This architectural pattern decouples the user interface, business logic, and data storage, which is essential for maintaining a high-performance system capable of handling real-time navigation and information requests.

The architecture consists of the following three layers:

**Presentation Layer** – This layer provides the user interface (UI) where administrators, staff, and tourists interact with the system. For tourists, it features a mobile-responsive dashboard for viewing resort galleries, checking rates, and accessing the interactive map. For administrators and staff, it includes management panels for updating landmark coordinates, managing facility information, and viewing system logs. This layer employs Role-Based Access Control (RBAC) to ensure that menus are customized by user type—restricting sensitive administrative functions to authorized personnel while providing a streamlined, simplified interface for tourists to find their way to the resort.

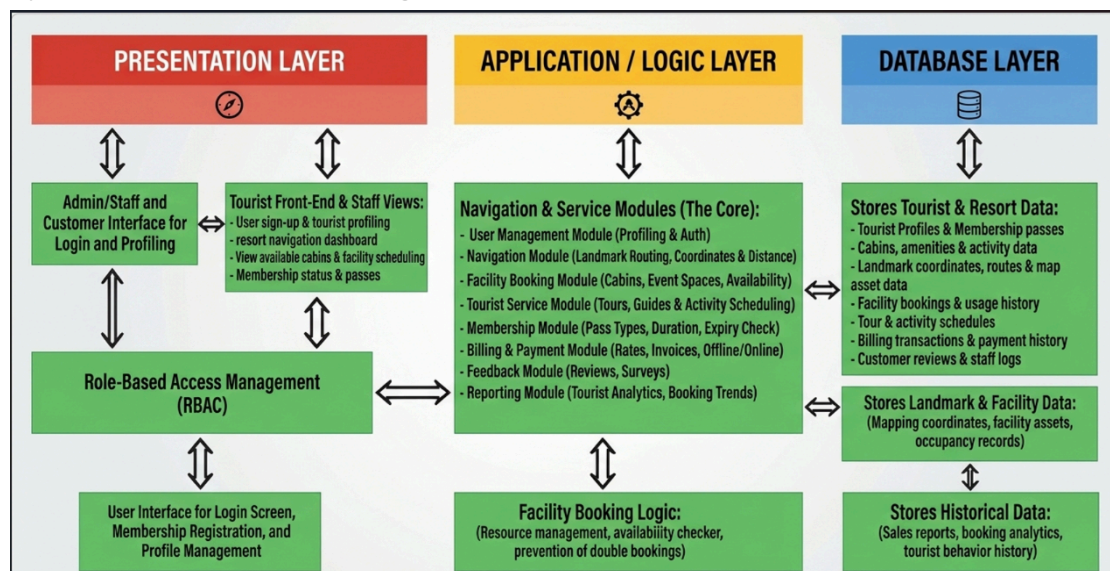
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**Application or Logic Layer** – Acting as the "brain" of the system, this layer processes all user requests and executes the core business logic. Key modules include the Navigation Module, which calculates routes based on specific landmarks and coordinates; the Information Management Module, which processes updates to resort amenities and pricing; and the Authentication Module, which validates login credentials for staff and admins. This layer ensures that all inputs

are validated and correctly processed (e.g., verifying map coordinates) before sending data to the database or returning results to the tourist's screen.

**Database Layer** – This layer is responsible for the secure and organized storage of all essential data using MySQL. It stores geographic coordinates for landmarks, high-resolution image paths, service rates, user account credentials, and system activity logs. This layer ensures data integrity and consistency, allowing for fast and reliable retrieval of information. This is critical for real-time navigation, as the system must pull precise landmark data and location details instantly to provide a smooth, dependable experience for the traveler.

### System Architecture Diagram



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### Tools and Technologies Used

- Frontend: HTML, CSS, JavaScript
- Backend: Node.js/Express.js

- Mapping Integration: OpenStreetMap
- Database: MySQL
- Development tools: Visual Studio Code

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**Data Gathering Techniques**

To design and develop the Online Information and Tourist Navigation System for KML Mountain Resort that effectively addresses the requirements of the management and the needs of travelers, the proponents will utilize specific data gathering methods. These techniques ensure that the system is technically sound,

navigationally accurate, and capable of solving the operational hurdles of the resort.

### **Surveys**

The proponents will distribute surveys to a target group of potential tourists and previous visitors of the resort. These surveys will focus on the "pain points" of travel, such as the difficulty of finding the location using existing maps, the clarity of landmark instructions, and the accessibility of service rates. The feedback will help the proponents determine which features—such as real-time coordinate tracking or high-resolution visual galleries—are most critical for the system's success. Questions will also evaluate the respondents' technological readiness to use a web-based navigation portal.

### **Interviews**

The proponents will conduct structured interviews with the resort owner and management staff. Since the staff currently handles inquiries and provides directions manually, these interviews are vital for capturing the "local knowledge" of the area. This includes identifying specific landmarks that are more reliable for navigation than standard GPS markers. The proponents will gather detailed information regarding the resort's operational flow, pricing structures, and the common navigational errors reported by guests, which will serve as the basis for the system's logic and routing modules.

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### **Site Observation and Field Mapping**

Unlike a standard management system, a navigation-based project requires physical verification. The proponents will visit the resort and the surrounding access roads to perform field mapping. This involves recording actual geographic coordinates at key turning points and landmarks. This firsthand data collection is

essential for ensuring that the Navigation Module provides precise, verified instructions that correspond to the actual terrain leading to KML Mountain Resort.

### **Internal Record Review**

Examining the resort's manual inquiry logs, guest manifestos, and existing price lists. By analyzing the most frequently asked questions and common navigational complaints recorded by staff, the proponents can prioritize specific features in the information hub.

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### **Testing Procedures**

Testing is a critical phase for the Online Information and Tourist Navigation System, ensuring that the platform operates as intended and meets all functional requirements before being used by tourists and resort staff.



- **Unit Testing:** This involves verifying individual components for correct functionality. In the KML system, this means testing specific features like the landmark coordinate calculator, the photo gallery carousel, and the administrative login form in isolation to ensure there are no bugs in the basic code.
- **Integration Testing:** This stage ensures that combined modules work together seamlessly. For example, proponents will test if the Database Layer correctly sends landmark data to the Navigation Module, and if that information is then accurately displayed on the Presentation Layer's interactive map.
- **System Testing:** This involves evaluating the complete system's performance under real-world conditions. For KML Mountain Resort, this includes "End-to-End" testing where the proponents simulate a tourist accessing the site from a mobile device on the road to verify that the navigation instructions and information hub load quickly and accurately in areas with varying signal strength.

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**Evaluation Criteria**

The success and effectiveness of the Online Information and Tourist Navigation System for KML Mountain Resort will be measured using the following criteria. These benchmarks ensure the system is not only functional but also provides high value to both the resort management and the visiting tourists.

### Accuracy and Reliability

- **Navigational Precision:** The primary measure of success is whether the landmark-based routing leads users to the resort without error. Success is defined by the alignment of system-provided coordinates with actual physical locations.
- **Data Integrity:** Information regarding service rates, room availability, and resort policies must be 100% consistent with the resort's official manual records.

### Usability and User Experience (UX)

- **Ease of Navigation:** The interface must be intuitive enough for first-time tourists to find the "Navigate" or "Rates" buttons within seconds of landing on the homepage.
- **Mobile Responsiveness:** Since most tourists will use the system while traveling, the platform must adjust seamlessly to various screen sizes (smartphones, tablets, and laptops).

### Performance and Efficiency

- **Load Times:** The interactive map and high-resolution image galleries must load quickly, even in areas where mobile data speeds might be limited (mountainous regions).
- **Information Retrieval:** The administrative dashboard must be able to generate reports or update landmark data instantly without system lag.

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### Security and Access Control

- **Data Protection:** Evaluation of how well the system prevents unauthorized access to the administrative backend. This includes testing the strength of the Role-Based Access Control (RBAC).
- **Input Validation:** The system must successfully block malicious inputs (such as SQL injection) in the inquiry forms or login screens.

### Scalability and Maintainability

- Future Updates: The system should be designed so that new landmarks, updated service rates, or additional resort features can be added by the staff without requiring a complete rewrite of the source code.
- System Stability: The ability of the web portal to handle multiple simultaneous visitors (e.g., during peak holiday seasons) without crashing.

### Evaluation Scale

The proponents will use a Likert-scale (typically 1 to 5) during User Acceptance Testing (UAT) to quantify these criteria:

| Rating           | Description                                                               |
|------------------|---------------------------------------------------------------------------|
| 5 - Excellent    | System exceeds all requirements and is highly intuitive.                  |
| 4 - Good         | System meets all requirements with minor suggestions for UI improvement.  |
| 3 - Fair         | System is functional but has occasional lag or requires a learning curve. |
| 2 - Poor         | System has significant bugs or is difficult for non-technical users.      |
| 1 - Unacceptable | System fails to provide accurate navigation or basic resort information.  |

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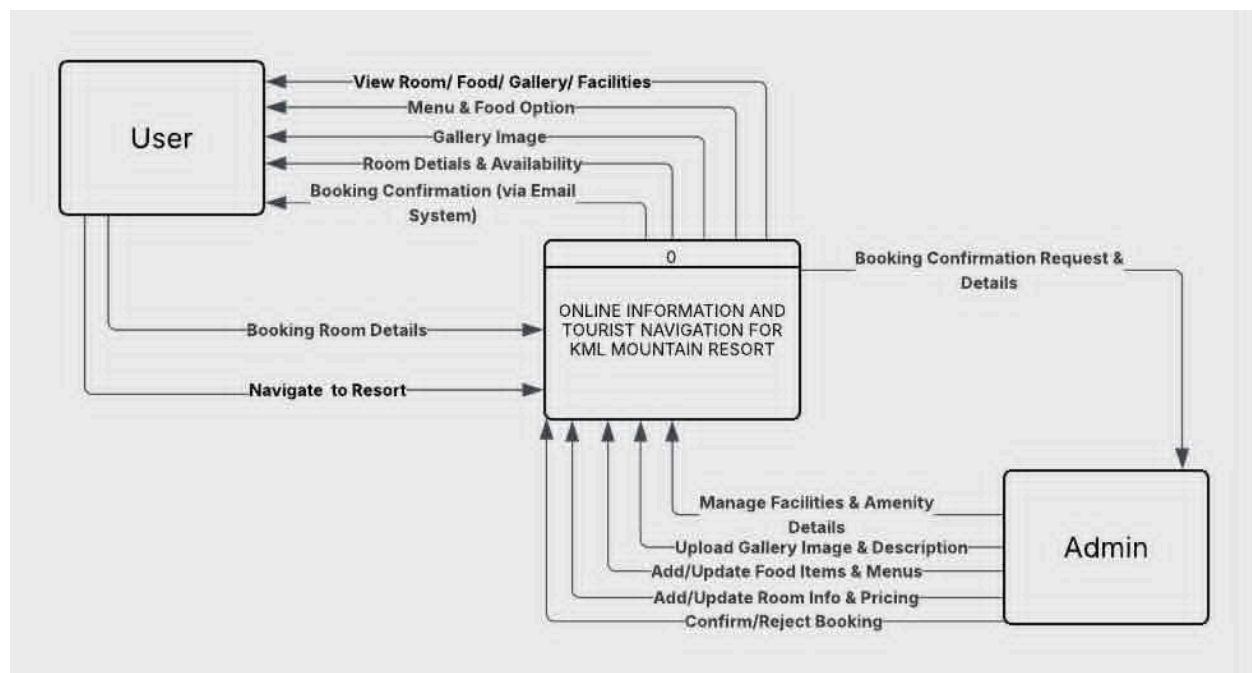
### SUPPORTING DIAGRAMS

#### DATA FLOW DIAGRAM (DFD)

The Data Flow Diagram illustrates the logical flow of information within the Online Information and Tourist Navigation System for KML Mountain Resort. It

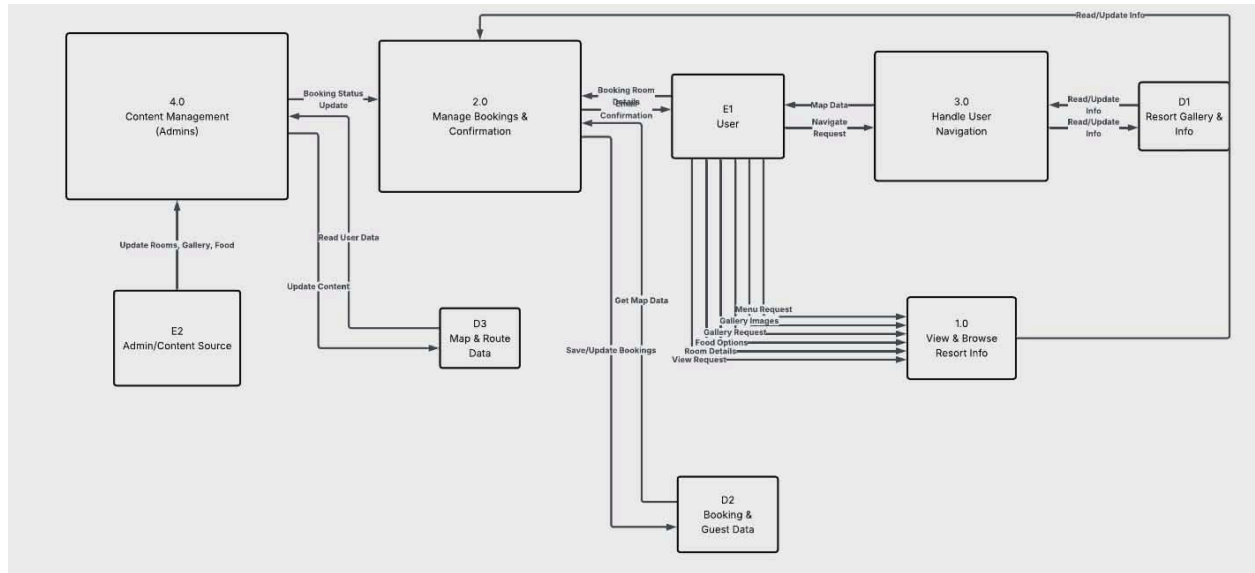
identifies the external entities (Users), processes (System Actions), data stores (Databases), and the specific data flows that move between them.

The Level 0 DFD provides a high-level overview of the entire system as a single process. It shows how information moves between the external entities and the KML MOUNTAIN RESORT System.



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The Level 1 DFD for the Online Information and Tourist Navigation System for KML Mountain Resort breaks down the high-level interactions into specific, functional sub-processes. This level reveals how data is transformed by the "brain" of the system the Application Layer before being stored or displayed.



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**ENTITY RELATIONSHIP DIAGRAM (ERD)**

The Entity Relationship Diagram presents the logical structure of the database used in the Online Information and Tourist Navigation System for KML Mountain

Resort. It defines the entities, their specific attributes, and the relationships that ensure data remains consistent and organized within the MySQL environment. The ERD for the KML system defines the relationships between the following key entities:

- Users (Admin, Staff, and Tourists): This central entity manages authentication and access levels. Each user is assigned a specific role which dictates what data they can modify or view.
- Tourist Profiles: Linked directly to the User entity, this stores personal details such as contact information and visitation history, allowing for a personalized experience.
- Resort Facilities / Cabins: Stores the details of the accommodations, including descriptions, high-resolution image paths (for the gallery), and current seasonal rates.
- Bookings & Reservations: This entity acts as the bridge between Tourists and Facilities. It tracks check-in/check-out dates, total billing amounts, and reservation status to prevent double bookings.
- Landmarks & Coordinates: A unique entity for this system that stores the geographical data (Latitude and Longitude) and descriptive markers used by the Navigation Module to generate routes.
- Inquiries & Feedback: Tracks communications between tourists and the resort staff, allowing for administrative monitoring of customer service quality.

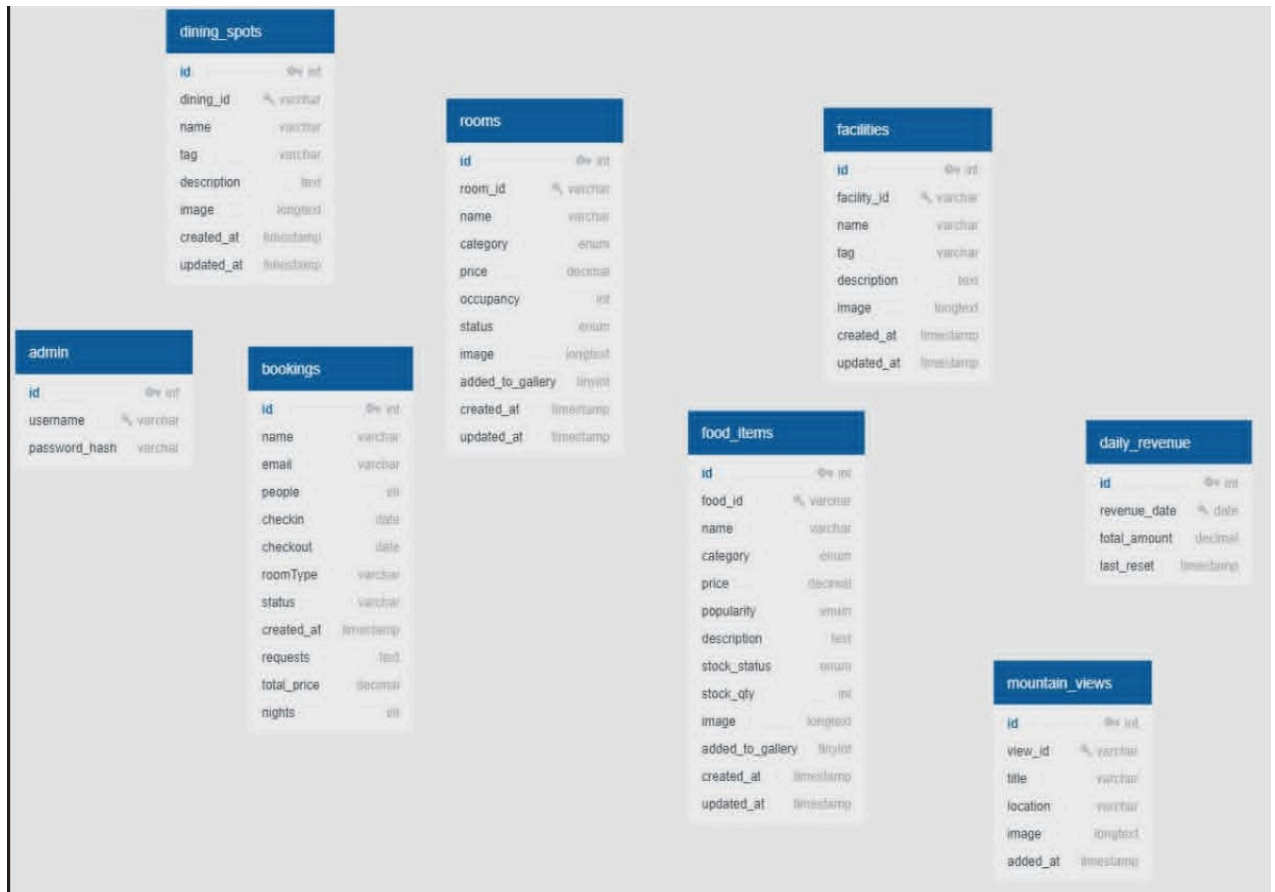
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#### Database Cardinality

The system utilizes standard relational logic to maintain order:

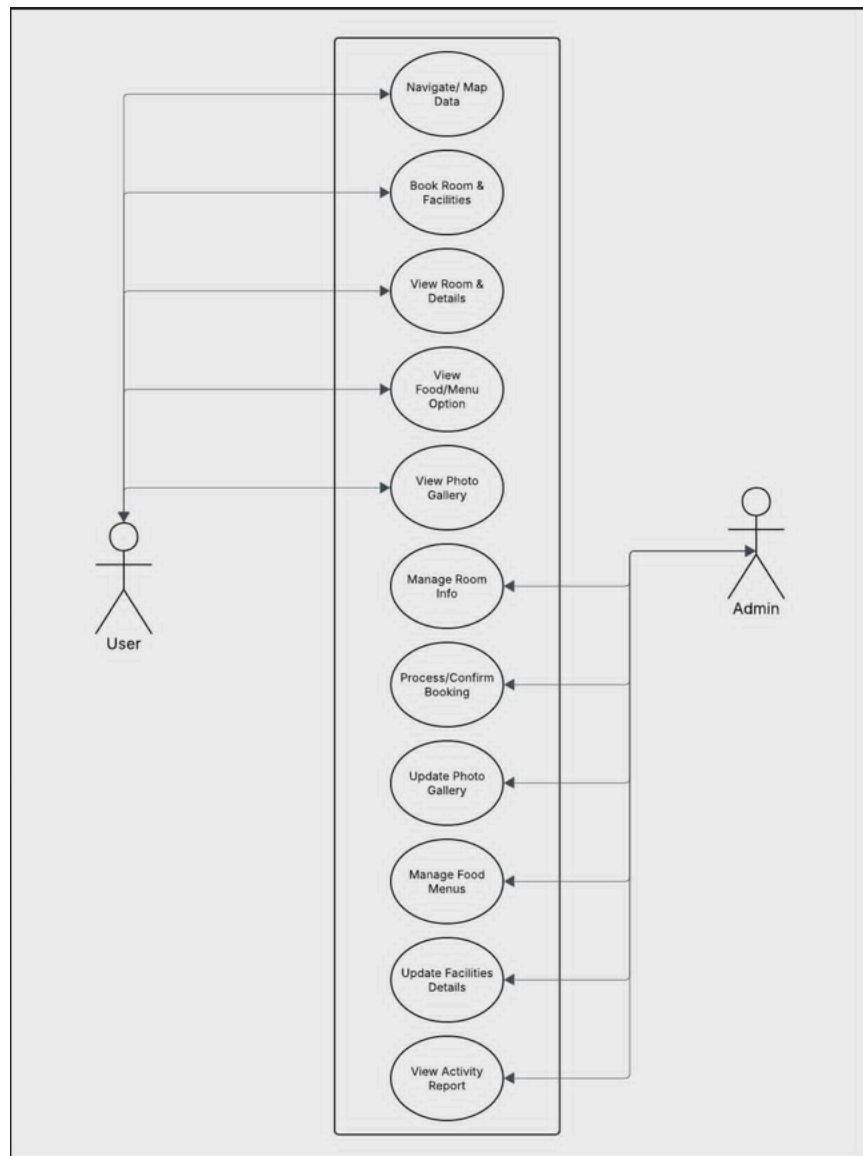
- One-to-Many: One Tourist can have multiple Bookings over time, but each booking is tied to a single tourist.

- One-to-Many: One Admin can manage and update multiple Landmarks and Facility records.
- Many-to-One: Multiple Inquiries can be directed toward a single Resort Facility.



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**USE CASE DIAGRAM**



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**Use-Case Diagram Analysis**



The Use-Case Diagram visually outlines the functional requirements of the Online Information and Tourist Navigation System. It defines the Actors and the System Functions, serving as a bridge between the resort's operational needs and the software's final design.

#### Primary Actors

- Tourist/Guest: The external user. Their primary goal is to consume information and reach the destination. They interact with the "front-facing" side of the system to simplify their travel experience.
- Administrator: Represented by the resort owner, Michelle Tingson Miranda. The admin maintains the system's accuracy, ensuring that the information provided to the guests is current and that business operations (like bookings) are flowing smoothly.

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#### Key Use-Cases and Their Significance

| Use Case                 | Description                                                                                                | Primary Actor |
|--------------------------|------------------------------------------------------------------------------------------------------------|---------------|
| View Resort Information  | Accessing the digital gallery, food menus, and room details to make an informed visit.                     | Tourist       |
| Inquire/Book Room        | Completing the digital form to initiate a reservation, moving away from manual, unrecorded inquiries.      | Tourist       |
| Access Live Navigation   | Utilizing the Landmark-based Interactive Map to receive real-time routing to the resort.                   | Tourist       |
| Manage Content           | The ability to upload new photos, update food options, or change facility descriptions.                    | Admin         |
| Approve/Process Bookings | Transitioning a guest request from "Pending" to "Confirmed" within the system dashboard.                   | Admin         |
| Monitor Analytics        | Reviewing activity reports to see which navigation landmarks are most used or which rooms are most viewed. | Admin         |

### System Boundary and Interactions

As shown in the provided diagram, the System Boundary (the rectangle) separates the internal functions of the KML platform from the external actors.

- Tourist Interactions: These are largely "read" and "submit" actions.
- Admin Interactions: These are "management" and "control" actions that dictate what is stored in the database.
- Automation: Some use cases, such as "View Photo Gallery," are purely informational, while "Process Booking" involves complex logic that updates the status of the resort's availability in real-time.